



SUMO NOTES

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Tutorials

- Installation with good explanation: [SUMO - Simulation of Urban Mobility installation and PATH configuration](#)
- Traci4Matlab installation: [SUMO TraCI4Matlab Installation Tutorial](#)
- A detailed tutorial for traCI: <https://intelaligent.github.io/tctb/post-learning-traci-tls.html#start-the-simulation-in-sumo-gui>
- Running test2. : The tutorials files might not run if you are in the installation directory. Copy the tutorial file to another directory (for example, **Documents> SUMO**)
- <https://stackoverflow.com/questions/67742345/how-to-use-runner-py-in-sumo-via-command-line-and-solving-its-common-errorpermi>
- <https://www.mail-archive.com/sumo-user@eclipse.org/msg08364.html>
- Follow the tutorial: <https://www.youtube.com/watch?v=258Y0e0u-Go>

Remind that, with newer SUMO versions, traci4matlab installs automatically and all the path and environ. Variables are created automatically during installation.

Procedure to install TraCI4Matlab:

- 1) Download SUMO
- 2) Download & Install TraCI4Matlab from GitHub (<https://github.com/pipeacosta/traci4matlab>)
- 3) My Computer > Properties > Advanced system settings > Environment variables > system variables > Path ----> C:\Program Files (x86)\Eclipse\Sumo\bin
- 4) My Computer > Properties > Advanced system settings > Environment variables > system variables > New ----> Variable Name : SUMO_HOME , Path of the variable : C:\Program Files (x86)\Eclipse\Sumo > ok
- 5) Create **javaclasspath** text file, containing C:\Program Files (x86)\Eclipse\Sumo\SUMO_HOME\tools\contributed\traci4matlab-master\traci4matlab.jar
- 6) Copy the text file to C:\Users\sk_chada\AppData\Roaming\MathWorks\MATLAB\R2019a
- 7) Goto Matlab > pathtool > add folder > C:\Program Files (x86)\Eclipse\Sumo\SUMO_HOME\tools\contributed\traci4matlab-master ----> save
- 8) Run **inter_palmas_actuated.m** to test if the path is correctly given

Do not forget Step 5 & 6. Without this following error might appear at Matlab

```
"Could not connect to TraCI server at 127.0.0.1:~. Error using fprintf
Unable to convert 'MException' value to 'char' or 'string'.

Error in traci.connect (line 47)
    fprintf('Could not connect to TraCI server at %s:%s. %s\n', host, port, err);

Error in traci.start (line 62)
    connections(label) = traci.connect(port, numRetries, host);

Error in traci_test (line 22)
    [traciVersion,sumoVersion] = traci.start(['sumo -c ' ~ scenarioPath ~']);"
```

Speed Modes

speed mode (0xb3)

This command controls how speeds set with the command *setSpeed* (0x40) and *slowDown* (0x14) are used. Per default, the vehicle may only drive slower than the speed that is deemed safe by the car following model and it may not exceed the bounds on acceleration and deceleration. Furthermore, vehicles follow the right-of-way rules when approaching an intersection and if necessary they brake hard to avoid driving across a red light. One can control this behavior using the speed mode (0xb3) command. The given integer is a bitset (bit0 is the least significant bit) with the following fields:

- bit0: Regard safe speed
- bit1: Regard maximum acceleration
- bit2: Regard maximum deceleration
- bit3: Regard right of way at intersections (only applies to approaching foe vehicles outside the intersection)
- bit4: Brake hard to avoid passing a red light
- bit5: **Disregard** right of way within intersections (only applies to foe vehicles that have entered the intersection).

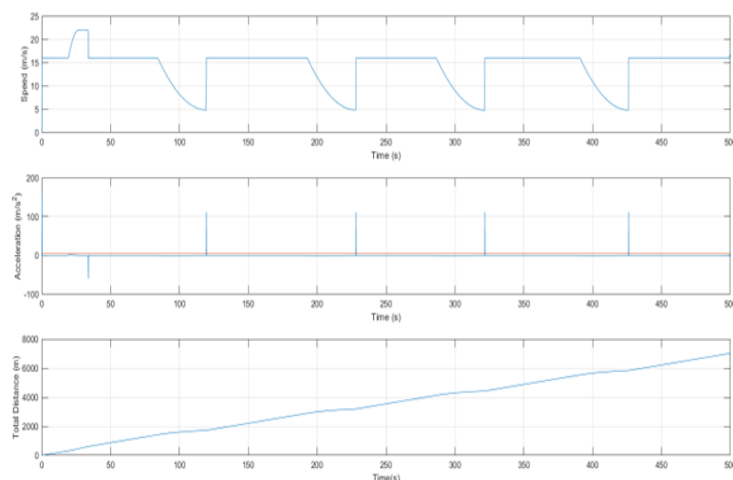
Setting the bit enables the check (the according value is regarded), keeping the bit=zero disables the check.

Examples:

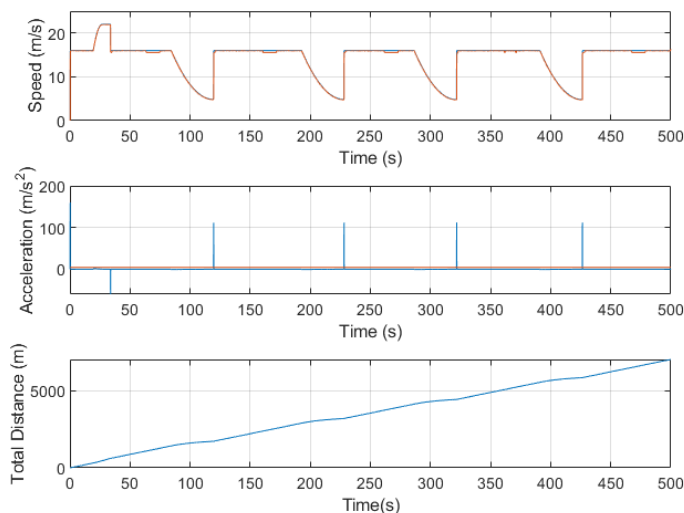
- default (all checks on) -> [0 1 1 1 1] -> Speed Mode = 31
- most checks off (legacy) -> [0 0 0 0 0] -> Speed Mode = 0
- all checks off -> [1 0 0 0 0] -> Speed Mode = 32
- disable right of way check -> [1 1 0 1 1] -> Speed Mode = 55
- run a red light [0 0 0 1 1] = 7 (also requires *setSpeed* or *slowDown*)
- run a red light even if the intersection is occupied [1 0 0 1 1] = 39 (also requires *setSpeed* or *slowDown*)

Caution

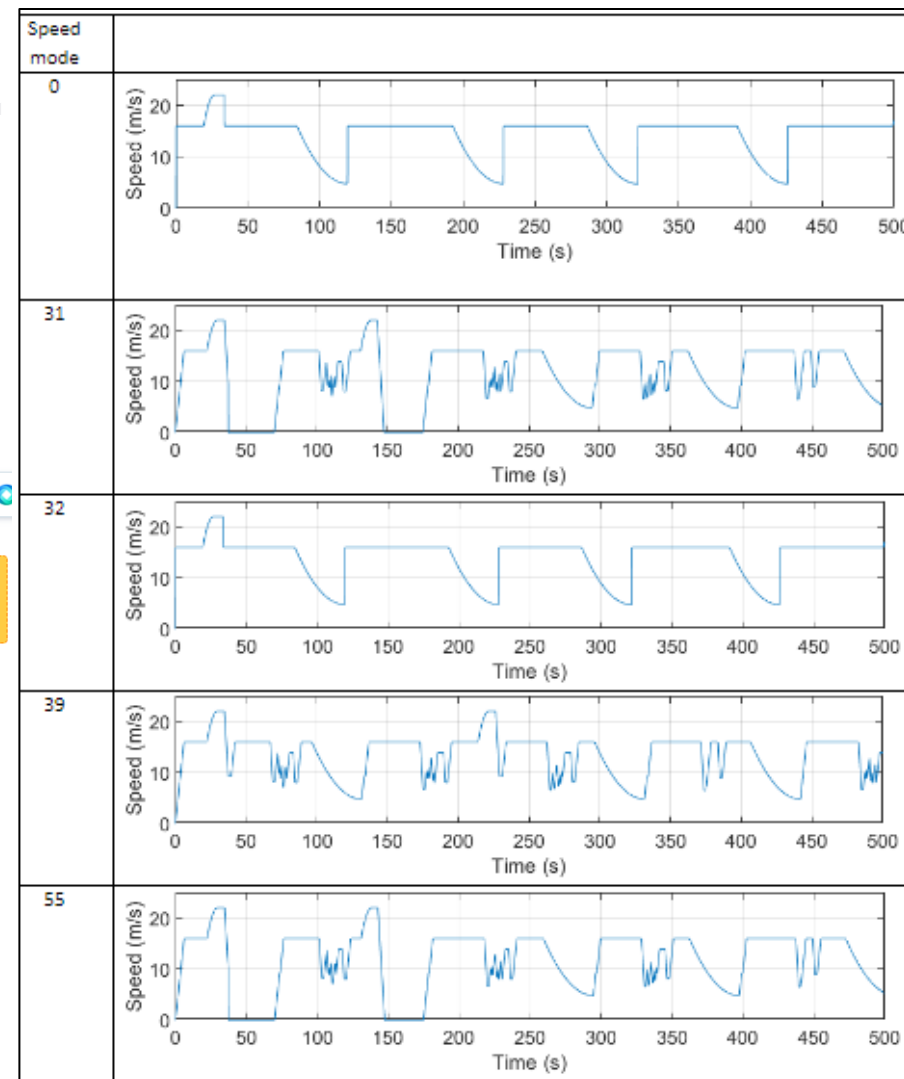
bit5 has inverted semantics and must be set to '1' in order to disable the safety function. This achieves backward compatibility with earlier versions of SUMO where this bit was not defined and right of way within intersection could not be ignored explicitly.



Speed mode 32



Interfering- speed mode 0



Simulation Loop

https://sumo.dlr.de/docs/Developer/Implementation_Notes/Simulation_Loop.html

Application order of traci commands and simulation step

Step **n**:

1. *traci.vehicle.getPosition* and *traci.vehicle.getSpeed* show the values from step **n-1**
2. when calling *traci.vehicle.moveTo* the vehicle is instantly placed (and still moves in this step)
3. *vNext* (the speed for this step) is computed based on the values of the previous step (*traci.vehicle.setSpeed* overrides this)
4. vehicle position is updated based on *vNext* (either by adding *vNext* or by adding $(vPrev + vNext) / 2$ depending on the value of option **--step-method.ballistic**)
5. the value given to *traci.vehicle.moveToXY* overrides the previously computed position
6. GUI is updated to show final state of step **n**, outputs like *fcd-output* also show this state for step **n**

Step **n+1**

This means, that the position given to *moveToXY* in step **n** will be visible in the outputs for step **n** whereas *traci.vehicle.getPosition* will always show the value from the previous step.

Simplify External control loop:

<https://github.com/eclipse/sumo/issues/11091>

How to get the next SUMO speed/ acceleration for a vehicle when it is manually controlled? #8661

<https://github.com/eclipse/sumo/issues/8661>

How velocity applies for a vehicle ?

<https://www.eclipse.org/lists/sumo-user/msg08500.html>

I thought SpeedFactor was a user-defined parameter with the default value of 1.0? What do you mean by it's randomly selected?

Does this mean the following?

The SpeedFactor we define in .rou file is the mean of the distribution and another SpeedFactor is randomly selected based on speedDev (let's call this random SpeedFactor).

The random SpeedFactor is different for every vehicle but the same distribution of this random SpeedFactor is applied to the fleet as a whole.

The random SpeedFactor is multiplied to the lane speed limit to determine the vehicle's desired speed.

Finally, the vehicle speed is determined by $\min(\text{maxSpeed}, \text{desired speed})$ where maxSpeed is defined in the .rou file.

Thank you!

From: sumo-user-bounces@xxxxxxxxx <sumo-user-bounces@xxxxxxxxx> on behalf of Jakob Erdmann <namdre.sumo@xxxxxxxxx>

Sent: Saturday, December 19, 2020 4:52 PM

To: Sumo project User discussions <sumo-user@xxxxxxxxx>

Subject: Re: [sumo-user] Understanding SpeedDev.

No. SpeedFactor is selected randomly at vehicle creation based on speedDev and remains constant over the vehicle lifetime (unless changed via traci or type calibrator). The speedFactor distribution thus applies to the fleet as a whole.

Am Sa., 19. Dez. 2020 um 18:14 Uhr schrieb Bae, Jong In <jbae42@xxxxxxxxx>:

Hello,

I asked this question a while ago but looking at it again, I was still confused if I'm understanding this correctly.

Let's assume the following to determine a single vehicle's speed.

Lane speed limit = 30 m/s

Speed Factor = 1.0

Speed Dev. = 0.1

First, the lane speed limit (30m/s) is multiplied by the speed factor (1.0).

And following the distribution and the speed dev (0.1), ~68% of the time, the vehicle will drive between 27m/s and 33m/s, ~95% of the time, the vehicle will drive 24m/s and 36m/s.

This determines the vehicle's desired speed.

Finally, the vehicle speed is $\min(\text{maxSpeed}, \text{desired speed})$ where maxSpeed is defined in the .rou file.

Could you please confirm if this is the right understanding?

Thank you!

Speed Factor:

<https://github.com/eclipse/sumo/issues/11407>

How to get the next SUMO speed/ acceleration for a vehicle when it is manually controlled? #8661

<https://github.com/eclipse/sumo/issues/8661>



If you are using traci interface, you can set any individual vehicle speed from the next timestep using the command

0



```
traci.vehicle.setSpeed(vehId, desiredspeed)
```

Vehicle Type

- VehicleType Defaults: https://sumo.dlr.de/docs/Vehicle_Type_Parameter_Defaults.html

vClass (SVC)	example	shape (SVS)	length width height	minGap	a _{max} ^(**) accel	b ^(**) decel	b _{emergency} decel	V _{max} maxSpeed	V _{desMax} desiredMaxSpeed	seats	emissionClass (HBEFA3)	speed deviation
pedestrian		pedestrian	0.215m ⁽¹⁾ 0.478m ⁽¹⁾ 1.719m ⁽¹⁾	0.25m	1.5m/s ²⁽²³⁾	2m/s ²⁽²³⁾	5m/s ²	37.58km/h ⁽⁴¹⁾	5km/h ⁽²³⁾	-	zero	0.1
two-wheeled vehicle												
bicycle		bicycle	1.6m ⁽¹⁷⁾ 0.65m ⁽¹⁷⁾ 1.7m ^(*)	0.5m	1.2m/s ²⁽¹⁹⁾	3m/s ²⁽¹⁹⁾	7m/s ²	50km/h	20km/h ⁽¹⁹⁾	1	zero	0.1
moped		moped	2.1m ⁽¹⁷⁾ 0.8m ⁽¹⁷⁾ 1.7m ^(*)	2.5m	1.1m/s ²⁽²⁵⁾	7m/s ²⁽²⁶⁾	10m/s ²	45km/h ⁽²⁴⁾	not limited (1e4km/h)	2	LDV_G_EU6	0.1
motorcycle		motorcycle	2.2m ⁽²⁸⁾ 0.9m ⁽²⁸⁾ 1.5m ⁽²⁸⁾	2.5m	6m/s ²⁽¹⁹⁾	10m/s ²⁽²⁷⁾	10m/s ²	200km/h ⁽²⁸⁾	not limited (1e4km/h)	2	LDV_G_EU6	0.1
passenger car												
passenger		passenger	5 ⁽²⁹⁾ 1.8m ⁽²⁹⁾ 1.5m ⁽²⁹⁾	2.5m	2.6m/s ²⁽²⁹⁾	4.5m/s ²⁽²⁷⁾	9m/s ²	200km/h ⁽²⁹⁾	not limited (1e4km/h)	4	PC_G_EU4	0.1
		passenger/sedan	see above	see above	see above	see above	see above	see above	see above	5	PC_G_EU4	0.1
		passenger/hatchback	see above	see above	see above	see above	see above	see above	see above	5	PC_G_EU4	0.1



Randomness

- How to add randomness ?

<https://sumo.dlr.de/docs/Simulation/Randomness.html>

- How to add randomness (example, randomness in vehicle DepartTime) ?

Departure times

The departure times of all vehicles may be varied randomly by using the option **--random-depart-offset** *<TIME>*. When this option is used each vehicle receives a random offset to its departure time, equidistributed on [0, *<TIME>*].

Detectors

- How to Add Detectors to the Network ?
- Use the default python script they provide: <https://sumo.dlr.de/docs/Tools/Output.html>
- Other videos available as well: <https://www.youtube.com/watch?v=VtLET4rdA7s>

Traffic Signals

- How to get phase duration of traffic signal ? : <https://www.eclipse.org/lists/sumo-user/msg00212.html>

First you need to determine the traffic-light index which corresponds to that edge (or rather to the lane-to-lane connections from that edge because each connection has it's own index). This is done by calling the function `trafficlights.getControlledLinks`.

Then you can retrieve the state with `getRedYellowGreenState` and interpret it using the respective index)

By calling `getDuration` `getNextSwitch` you can figure out all there is to know about spend and remaining duration (see the explanation at `getPhaseDuration`)

Traci Start Configuration

- `traci.start([])` configuration options: <https://sumo.dlr.de/docs/sumo.html>

Loading order of input files

Sumo loads its input files in the following order

- net-file
- additional-files
- weight-files
- state-files
- route-files

Whenever a simulation object A (such as a vehicle) defined in an input file refers to another simulation object B (such as a vehicle type), the corresponding object B must be defined before A either by being put into a file that is loaded earlier, or by being put into the same file on an earlier line.

This is of particular importance in regard to files loaded with the options `--route-files <FILE>` and `--additional-files <FILE>`. The most important feature of the option `--route-files <FILE>` is that it loads the files incrementally thus saving memory. However, this means vehicles and flows must be sorted by their `departure` and `begin` times.

The option `--additional-files <FILE>` is handled before option `--route-files <FILE>` and the given files are fully loaded before the simulation starts. All items which can be put into a *route-file* can also be put into an *additional-file* (no vehicle sorting is necessary). Some objects must be available at the start of the simulation (such as output objects) and thus need to be loaded with this option, but is also useful for loading things referenced by route files such as `<vType>` -elements, named `<route>` elements or `<routeDistribution>`.

When multiple files are given as a comma-separated list, they are loaded from left to right.

Edge Lane Emission

<https://sumo.dlr.de/docs/Simulation/Output/Lane- or Edge-based Emissions Measures.html>

Caution

Please note the the unit of fuel-related outputs changed with SUMO 1.14.0 from liters to milligram. For the old behavior use the option **--emissions.volumetric-fuel**.

CO_abs	mg	The complete amount of CO emitted by the vehicles on this lane/edge during the aggregation period
CO2_abs	mg	The complete amount of CO ₂ emitted by the vehicles on this lane/edge during the aggregation period
HC_abs	mg	The complete amount of HC emitted by the vehicles on this lane/edge during the aggregation period
PMx_abs	mg	The complete amount of PM _x emitted by the vehicles on this lane/edge during the aggregation period
NOx_abs	mg	The complete amount of NO _x emitted by the vehicles on this lane/edge during the aggregation period
fuel_abs	mg	The complete amount of fuel the vehicles on this lane/edge during the aggregation period have consumed
electricity_abs	Wh	The complete amount of electricity the vehicles on this lane/edge during the aggregation period have consumed

CO_perVeh	mg	Assumed CO emissions a vehicle would produce when passing the edge
CO2_perVeh	mg	Assumed CO ₂ emissions a vehicle would produce when passing the edge
HC_perVeh	mg	Assumed HC emissions a vehicle would produce when passing the edge
PMx_perVeh	mg	Assumed PM _x emissions a vehicle would produce when passing the edge
NOx_perVeh	mg	Assumed NO _x emissions a vehicle would produce when passing the edge
fuel_perVeh	ml	Assumed fuel consumption a vehicle would need when passing the edge
electricity_perVeh	Wh	Assumed electricity consumption a vehicle would need when passing the edge

CO_normed	g/km/h	CO emissions during this interval normed by time and edge/lane length
CO2_normed	g/km/h	CO ₂ emissions during this interval normed by time and edge/lane length
HC_normed	g/km/h	HC emissions during this interval normed by time and edge/lane length
PMx_normed	g/km/h	PM _x emissions during this interval normed by time and edge/lane length
NOx_normed	g/km/h	NO _x emissions during this interval normed by time and edge/lane length
fuel_normed	l/km/h	fuel consumption during this interval normed by time and edge/lane length
electricity_normed	W/km	electricity consumption during this interval normed by time and edge/lane length

Discussion

Omitting empty Edges/Lanes

17.07.2009: I have excluded computation of mean per-vehicle emission to pass a lane/edge though almost the same functionality is already covered by the "excludeEmpty"-attribute. I am not sure whether this makes sense.

Also, we maybe should then also decide whether or not this also should be done for travel time computation.

Value Range

17.07.2009: The values generated for absolute (..._abs) emission during the period and for the mean per-vehicle emission to pass a lane/edge are now given in mg for making lower values visible though only digits after the dot are used. I am not really happy with the results. If a vehicle passes a lane of 390m length at 10m/s, it is assumed it emits less CO than 0.005mg so it does not appear in the outputs; On the other hand, if one looks at a jammed situation, the emissions of CO₂ may climb up to CO_perVeh="21246834688.00" (sure, they would get infinite if the vehicle would be really standing). I suppose we have to evaluate how the system behaves.

sampledSeconds	s	The number of vehicles that are present on the edge/lane in each second summed up over the measurement interval (may be subseconds if a vehicle enters/leaves the edge/lane).
traveltime	s	Time needed to pass the edge/lane, note that this is just an estimation based on the mean speed, not the exact time the vehicles needed. The value is based on the time needed for the front of the vehicle to pass the edge.
overlapTraveltime	s	Time needed to pass the edge/lane completely, note that this is just an estimation based on the mean speed, not the exact time the vehicles needed. The value is based on the time any part of the vehicle was the edge.

density	#veh/km	Vehicle density on the edge
laneDensity	#veh/km/lan	Vehicle density on the edge per lane

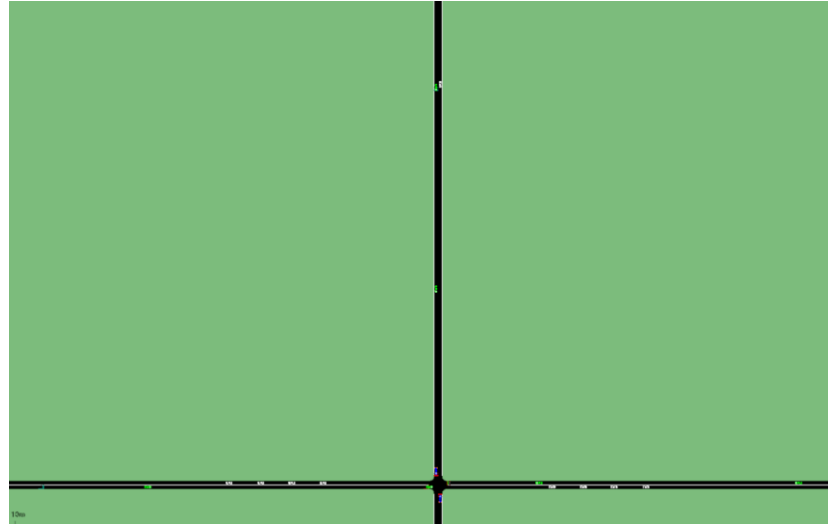
waitingTime	s	The total number of seconds vehicles were considered halting (speed < speedThreshold). Summed up over all vehicles
timeLoss	s	The total number of seconds vehicles lost due to driving slower than desired (summed up over all vehicles)
speed	m/s	The mean speed on the edge/lane within the reported interval. Caution: This is an average over time and space (space-mean-speed), rather than a local average over the vehicles (time-mean-speed). Since slow vehicles spend more time on the edge they will have a proportionally bigger influence on average speed.

Bugs in TRACI₄MATLAB

- How to fix the `traci.trafficlights.getNextSwitch` bug:
 - <https://github.com/pipeacosta/traci4matlab/issues/8>
- invalid speed returned by `getSpeedWithoutTraCI`
 - #10947 : <https://github.com/eclipse/sumo/issues/10947>

ERRORS

- vehID = fNS.AV.266
- VPH=400;
- MPR =100,
- Distance =300m



Output argument "t" (and possibly others) not assigned a value in the execution with "ECO_AND" function.

Error in `OC_SUMO` (line 32)

```
[t,u,x,y,J_Th,Jt,Ju]= ECO_AND(rho,tg,tr,t0,sgr,L,vm,v0,vM,am,aM,sys);
```

Error in `main_PR` (line 155)

```
[tempVel,OCacce{ii,i}] = OC_SUMO (sgr,tlsDistance,v_step,tg,tr); % adjusting tg-1 & tr
```

Fixed it by making '`--random-depart-offset 0`' ... in `traci.start`

vehicle:fNS.AV.266 Parameter		
Name	Value	Dynamic
lane [id]	gneE1_0	✓
position [m]	893.19	🖼
lateral offset [m]	0.00	🖼
speed [m/s]	21.00	🖼
lateral speed [m/s]	0.00	🖼
acceleration [m/s^2]	0.00	🖼
angle [degree]	180.00	🖼
slope [degree]	0.00	🖼
speed factor	1.00	🖼
time gap on lane [s]	7.24	🖼
waiting time [s]	0.00	🖼
ting time (accumulated, 100.00s) [s]	0.00	🖼
time since startup [s]	86442.30	🖼
time loss [s]	6.20	🖼
impatience	0.00	🖼
last lane change [s]	0.00	🖼
desired depart [s]	2394.80	✗
depart delay [s]	0.00	✗
odometer [m]	888.09	🖼
remaining [#]	11110846	✗
insertion period [s]	9.00	✗
stop info		✓
line		✗
CO2 [mg/s]	3361.82	🖼
CO [mg/s]	0.00	🖼
HC [mg/s]	0.05	🖼
NOx [mg/s]	1.01	🖼
PMx [mg/s]	0.04	🖼
fuel [mg/s]	1072.27	🖼
electricity [Wh/s]	0.00	🖼
noise (Harmonoise) [dB]	69.15	🖼
devices	tripinfo	✗
persons	0	🖼
containers	0	🖼
lcState right	unknown	✓
lcState left	unknown	✓

- Covert xml2csv online:
- <https://www.convertcsv.com/xml-to-csv.htm>
- Merge multiple csv together online: <https://merge-csv.com/>
- <https://products.aspose.app/cells/merger>